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## **WEEK 5 - ReactJS Hands ON**

### **Exercise 1**

Create a new React Application with the name “myfirstreact”, Run the application to print “welcome to the first session of React” as heading of that page.

1. To create a new React app, Install Nodejs and Npm from the following link:

<https://nodejs.org/en/download/>

2. Install Create-react-app by running the following command in the command prompt:

```
C:>npm install -g create-react-app
```

3. To create a React Application with the name of “myfirstreact”, type the following command:

```
C:>npx create-react-app myfirstreact
```

4. Once the App is created, navigate into the folder of myfirstreact by typing the following command:

```
C:>cd myfirstreact
```

5. Open the folder of myfirstreact in Visual Studio Code
6. Open the App.js file in Src Folder of myfirstreact
7. Remove the current content of “App.js”
8. Replace it with the following:

```
function App() {  
  return (  
    <h1> Welcome the first session of React </h1>  
  );  
}
```

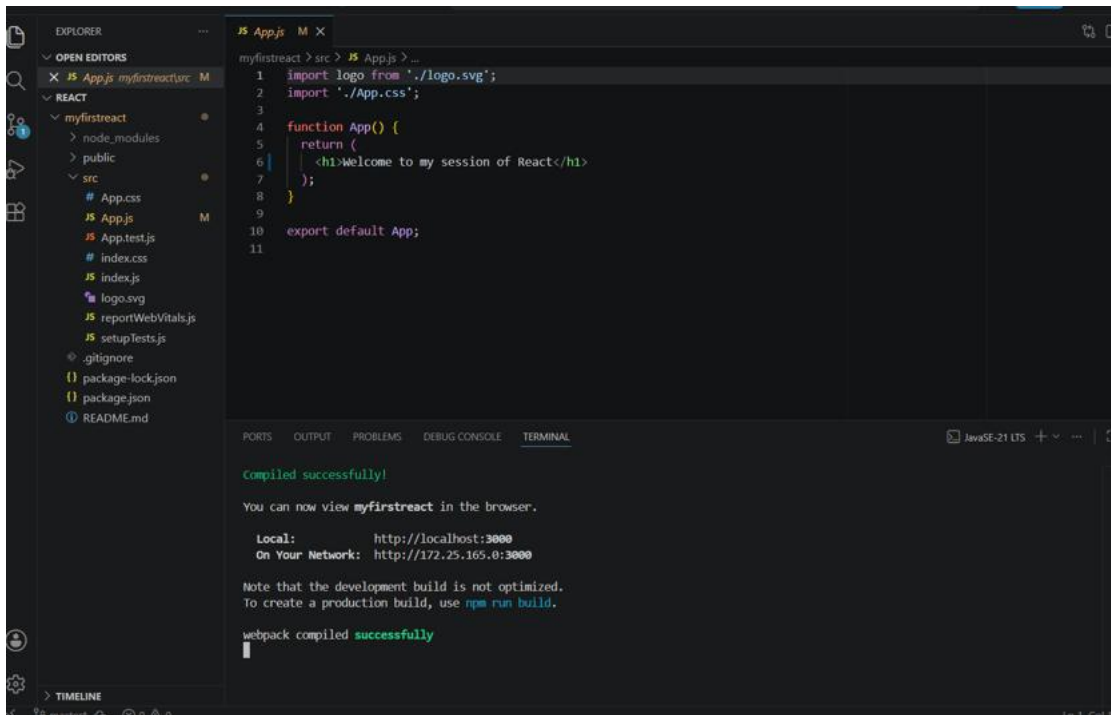
9. Run the following command to execute the React application:

```
C:\myfirstreact>npm start
```

10. Open a new browser window and type "localhost:3000" in the address bar



**Solution :**



**Welcome to my session of React**

## Exercise 2 :

Create a react app for Student Management Portal named StudentApp and create a component named Home which will display the Message “Welcome to the Home page of Student Management Portal”. Create another component named About and display the Message “Welcome to the About page of the Student Management Portal”. Create a third component named Contact and display the Message “Welcome to the Contact page of the Student Management Portal”. Call all the three components.

1. Create a React project named “StudentApp” type the following command in terminal of Visual studio:

```
C:>npx create-react-app StudentApp
```

2. Create a new folder under Src folder with the name “Components”. Add a new file named “Home.js”
3. Type the following code in Home.js

```
import React, {Component} from 'react';

import class Home extends Component{
  render(){
    <div>
      <h3> Welcome to the Home Page of Student Management Portal </h3>
    </div>
  }
}
```

4. Under Src folder add another file named “About.js”
5. Repeat the same steps for Creating “About” and “Contact” component by adding a new file as “About.js”, “Contact.js” under “Src” folder and edit the code as mentioned for “Home” Component.
6. Edit the App.js to invoke the Home, About and Contact component as follows:

```

import logo from './logo.svg';
import './App.css';
import {Home} from './Components/Home';
import {About} from './Components/About';
import {Contact} from './Components/Contact';

function App() {
  return (
    <div className="container">
      <Home/>
      <About/>
      <Contact/>
    </div>
  );
}

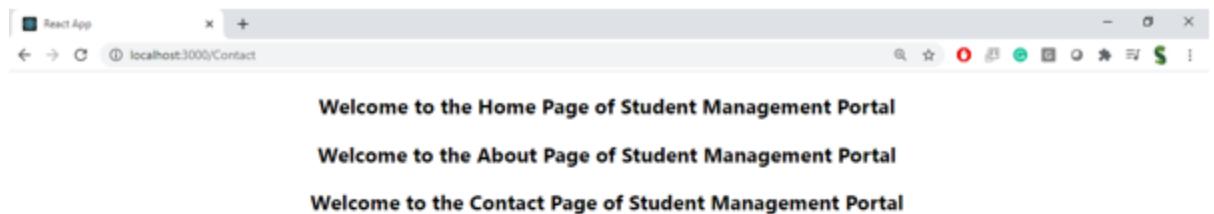
export default App;

```

7. In command Prompt, navigate into StudentApp and execute the code by typing the following command:

**C:\studentapp>npm start**

8. Open browser and type “localhost:3000” in the address bar:

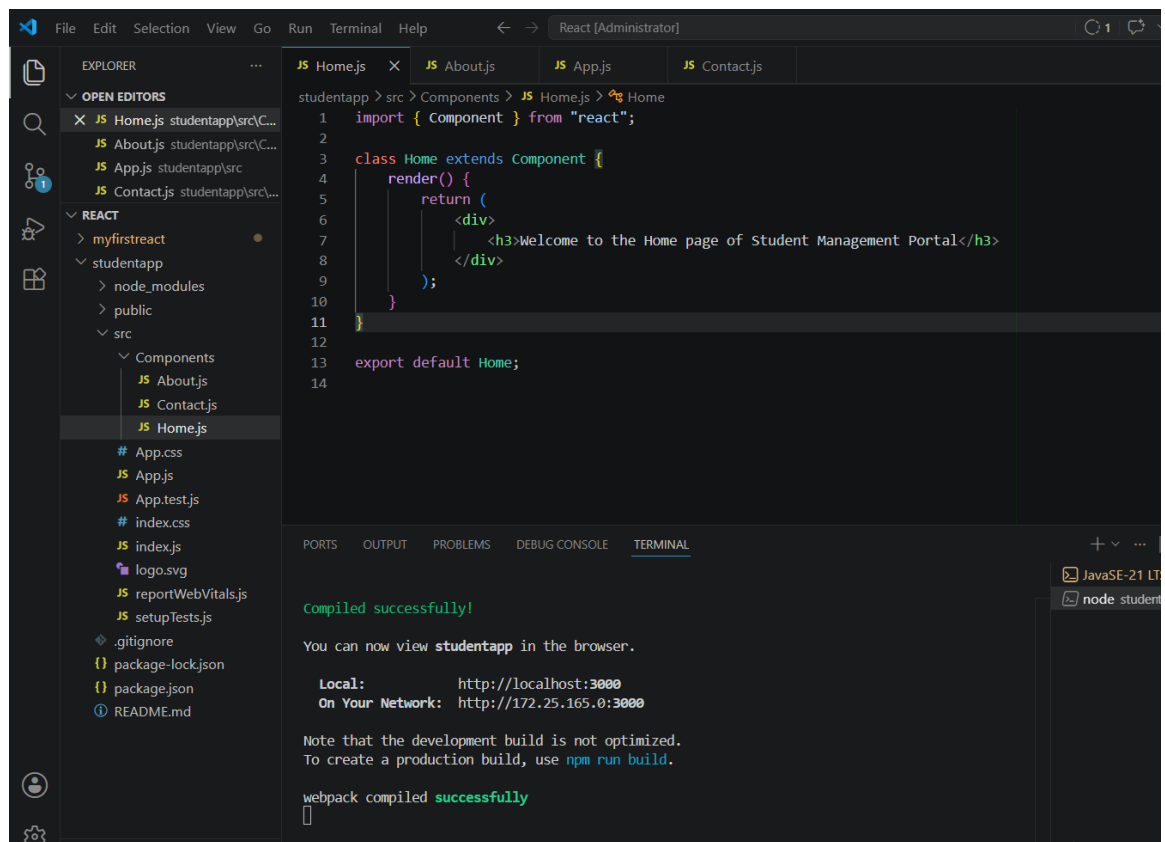


**Solution :**

## Home.js

```
import { Component } from "react";
```

```
class Home extends Component {  
  render() {  
    return (  
      <div>  
        <h3>Welcome to the Home page of Student Management Portal</h3>  
      </div>  
    );  
  }  
}
```



```
export default Home;
```

## About.js

```
import { Component } from "react";
```

```
class About extends Component {  
  render() {  
    return (  

```

```

    <div>
      <h3>Welcome to the About page of Student Management Portal</h3>
    </div>
  );
}
}

```

The screenshot shows a code editor with four tabs: JS Home.js, JS About.js (active), JS App.js, and JS Contact.js. The breadcrumb path is 'studentapp > src > Components > JS About.js > ...'. The code in About.js is as follows:

```

1  import { Component } from "react";
2
3  class About extends Component {
4    render() {
5      return (
6        <div>
7          <h3>Welcome to the About page of Student Manage
8        </div>
9      );
10   }
11 }
12
13 export default About;
14

```

```
export default About;
```

### Contact.js

```

import { Component } from "react";

class Contact extends Component {
  render() {
    return (
      <div>
        <h3>Welcome to the Contact page of Student Management
        Portal</h3>
      </div>
    );
  }
}

export default Contact;

```

```
JS Home.js JS About.js JS App.js JS Contact.js X
studentapp > src > Components > JS Contact.js > default
1  import { Component } from "react";
2
3  class Contact extends Component {
4    render() {
5      return (
6        <div>
7          <h3>Welcome to the Contact page of Student Management Portal</h3>
8        </div>
9      );
10   }
11 }
12
13 export default Contact;
```

## App.js

```
JS Home.js JS About.js JS App.js X JS Contact.js
studentapp > src > JS App.js > ...
6
7  function App() {
8    return (
9      <div className="App">
10        <Home />
11        <About />
12        <Contact />
13      </div>
14    );
15  }
16
17  export default App;
18
```



**Exercise 3 :** Create a react app for Student Management Portal named scorecalculatorapp and create a function component named “CalculateScore” which will accept Name, School, Total and goal in order to calculate the average score of a student and display the same.

1. Create a React project named “scorecalculatorapp” type the following command in terminal of Visual studio:

```
C:>npx create-react-app scorecalculatorapp
```

2. Create a new folder under Src folder with the name “Components”. Add a new file named “CalculateScore.js”
3. Type the following code in CalculateScore.js

```
import './Stylesheets/mystyle.css'

const percentToDecimal= (decimal) => {
  return (decimal.toFixed(2) + '%')
}

const calcScore = (total, goal) => {
  return percentToDecimal(total/goal)
}
```

```

export const CalculateScore = ({Name, School, total, goal}) => (
  <div className="formatstyle">
    <h1><font color="Brown">Student Details:</font></h1>
    <div className="Name">
      <b> <span> Name: </span> </b>
      <span>{Name}</span>
    </div>
    <div className="School">
      <b> <span> School: </span> </b>
      <span>{School}</span>
    </div>
    <div className="Total">
      <b><span>Total: </span> </b>
      <span>{total}</span>
      <span>Marks</span>
    </div>
    <div className="Score">
      <b>Score:</b>
      <span>
        {calcScore(
          total,
          goal
        )}
      </span>
    </div>
  </div>
)

```

4. Create a Folder named Stylesheets and add a file named "mystyle.css" in order to add some styles to the components:

```

.Name
{
  font-weight:300;
  color:blue;
}
.School
{
  color:crimson;
}
.Total
{
  color:darkmagenta;
}
.formatstyle
{
  text-align:center;
  font-size:large;
}
.Score
{
  color:forestgreen;
}

```

5. Edit the App.js to invoke the CalculateScore functional component as follows:

```

import {CalculateScore} from '../src/components/CalculateScore';

function App()
{
  return(
    <div>
      <CalculateScore Name={"Steeve"}
        School={"DNV Public School"}
        total={284}
        goal={3}
      />
    </div>
  )
}

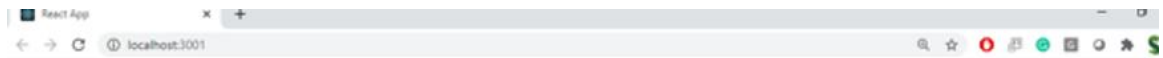
export default App;

```

6. In command Prompt, navigate into scorecalculatorapp and execute the code by typing the following command:

```
C:\scorecalculatorapp>npm start
```

7. Open browser and type "localhost:3000" in the address bar:



## Student Details:

Name: Steeve

School: DNV Public School

Total: 284Marks

Score:94.67%

### Solution :

CalculateScore.js

```
import "../Stylesheets/mystyle.css";
```

```
const percentToDecimal = (decimal) => {  
  return (decimal.toFixed(2) + "%");  
}
```

```
const calcScore = (total, goal) => {  
  return percentToDecimal((total / goal));  
}
```

```
export const CalculateScore = ({ Name, School, total, goal }) => {
```

```

<div className="formatstyle">

  <h1><font color="Brown">Student Details:</font></h1>


  <div className="Name">

    <b><span>Name:</span></b>

    <span>{Name}</span>

  </div>


  <div className="School">

    <b><span>School:</span></b>

    <span>{School}</span>

  </div>


  <div className="Total">

    <b><span>Total:</span></b>

    <span>{total}</span>

    <span> Marks</span>

  </div>


  <div className="Score">

    <b>Score:</b>

    <span>{calcScore(total, goal)}</span>

  </div>

</div>

);

```

```
JS CalculateScore.js U X # mystyle.css U JS App.js M JS index.js
scorecalculatorapp > src > components > JS CalculateScore.js > ...
1 import "../stylesheets/mystyle.css";
2
3 const percentToDecimal = (decimal) => {
4   return (decimal.toFixed(2) + "%");
5 }
6
7 const calcScore = (total, goal) => {
8   return percentToDecimal((total / goal));
9 }
10
11 export const CalculateScore = ({ Name, School, total, goal }) => (
12   <div className="formatstyle">
13     <h1><font color="Brown">Student Details:</font></h1>
14
15     <div className="Name">
16       <b><span>Name:</span></b>
17       <span>{Name}</span>
18     </div>
19
20     <div className="School">
21       <b><span>School:</span></b>
22       <span>{School}</span>
23     </div>
24
25     <div className="Total">
26       <b><span>Total:</span></b>
27       <span>{total}</span>
28       <span> Marks</span>
29     </div>
30
31     <div className="Score">
32       <b>Score:</b>
33       <span>{calcScore(total, goal)}</span>
34     </div>
35   </div>
36 );
```

Mystyle.css

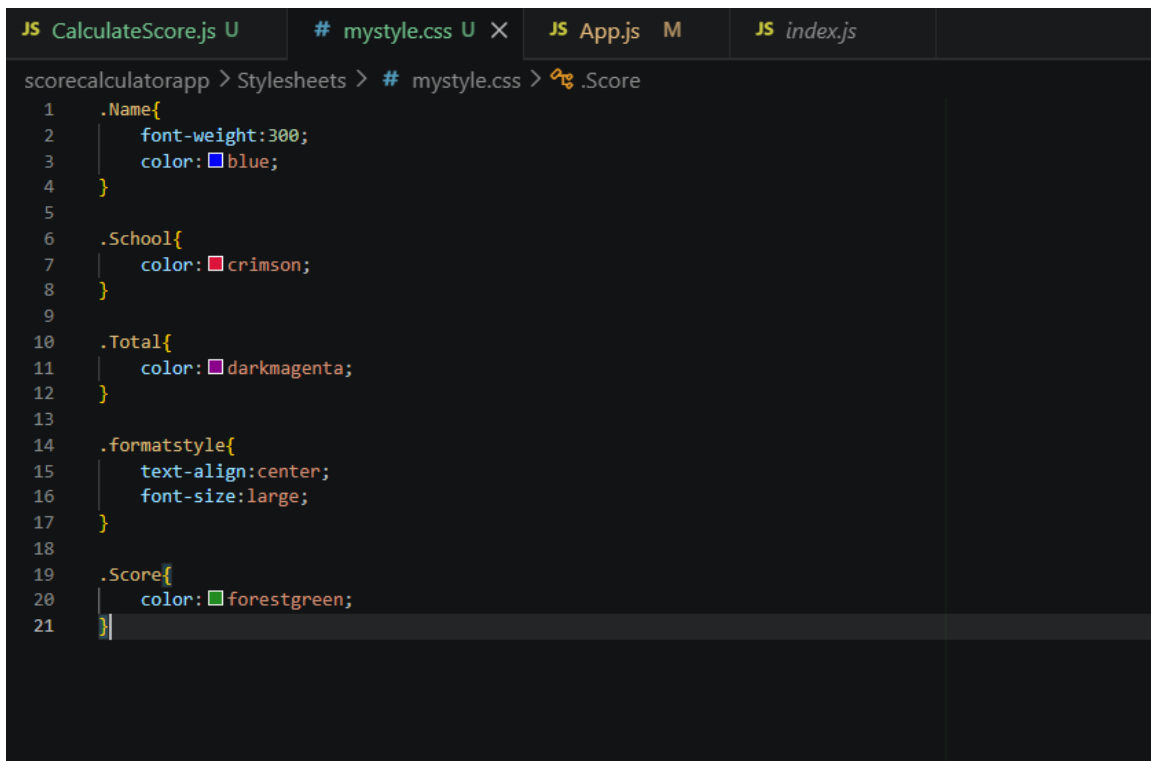
```
.Name{
  font-weight:300;
  color:blue;
}
```

```
.School{
  color:crimson;
}
```

```
.Total{
  color:darkmagenta;
}
```

```
.formatstyle{
    text-align:center;
    font-size:large;
}
```

```
.Score{
    color:forestgreen;
}
```



The screenshot shows a code editor with four tabs: 'CalculateScore.js U', '# mystyle.css U X', 'JS App.js M', and 'JS index.js'. The active tab is '# mystyle.css', which displays the following CSS code:

```
scorecalculatorapp > Stylesheets > # mystyle.css > .Score
1  .Name{
2      font-weight:300;
3      color:blue;
4  }
5
6  .School{
7      color:crimson;
8  }
9
10 .Total{
11     color:darkmagenta;
12 }
13
14 .formatstyle{
15     text-align:center;
16     font-size:large;
17 }
18
19 .Score{
20     color:forestgreen;
21 }
```

App.js

```
import { CalculateScore } from "../Components/CalculateScore";
```

```
function App() {
    return (
        <div>
```

```

    <CalculateScore
      Name={"Steve"}
      School={"DNV Public School"}
      total={284}
      goal={3}
    />
  </div>
);
}

```

```
export default App;
```

```

scorecalculatorapp > src > JS App.js > default
1  import { CalculateScore } from "../components/CalculateScore";
2
3  function App() {
4    return (
5      <div>
6        <CalculateScore
7          Name={"Steve"}
8          School={"DNV Public School"}
9          total={284}
10         goal={3}
11        />
12      </div>
13    );
14  }
15
16  export default App;

```

Result :





## Student Details:

**Name:** Steve

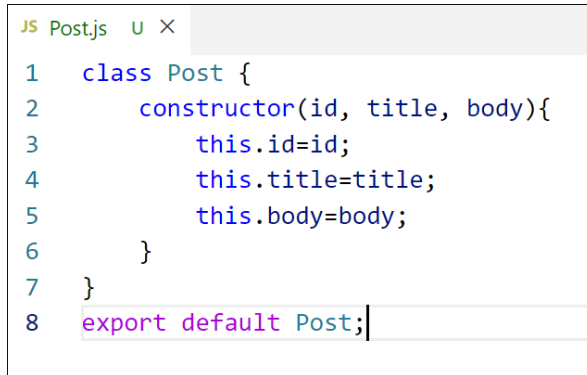
**School:** DNV Public School

**Total:** 284Marks

**Score:** 94.67%

#### Exercise 4 :

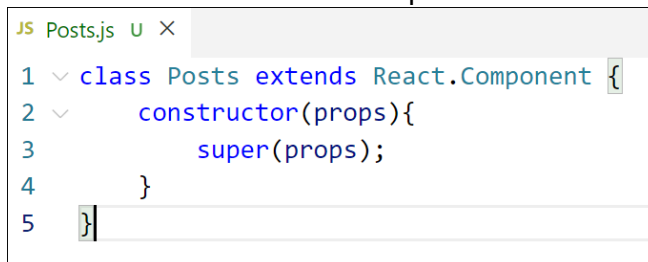
1. Create a new react application using *create-react-app* tool with the name as "blogapp"
2. Open the application using VS Code
3. Create a new file named as **Post.js** in **src folder** with following properties

A screenshot of a VS Code editor window with a tab labeled 'JS Post.js'. The editor contains the following code:

```
1 class Post {
2   constructor(id, title, body){
3     this.id=id;
4     this.title=title;
5     this.body=body;
6   }
7 }
8 export default Post;
```

Figure 1: Post class

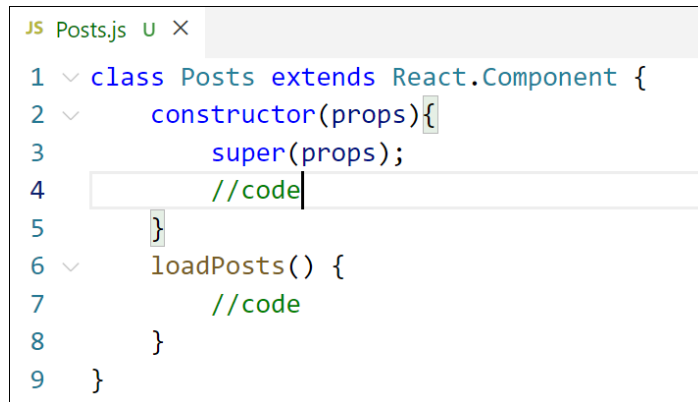
4. Create a new class based component named as **Posts** inside **Posts.js** file

A screenshot of a VS Code editor window with a tab labeled 'JS Posts.js'. The editor contains the following code:

```
1 class Posts extends React.Component {
2   constructor(props){
3     super(props);
4   }
5 }
```

Figure 2: Posts Component

5. Initialize the component with a list of Post in state of the component using the constructor
6. Create a new method in component with the name as **loadPosts()** which will be responsible for using Fetch API and assign it to the component state created earlier. To get the posts use the url (<https://jsonplaceholder.typicode.com/posts>)



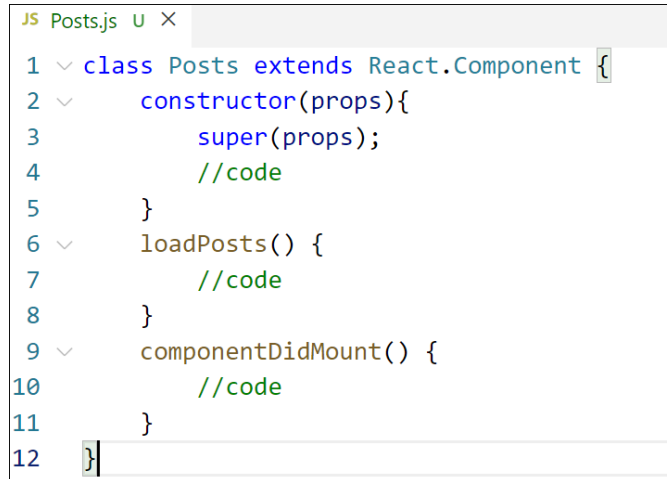
```

JS Posts.js U X
1 class Posts extends React.Component {
2   constructor(props){
3     super(props);
4     //code
5   }
6   loadPosts() {
7     //code
8   }
9 }

```

**Figure 3: loadPosts() method**

7. Implement the **componentDidMount()** hook to make calls to **loadPosts()** which will fetch the posts



```

JS Posts.js U X
1 class Posts extends React.Component {
2   constructor(props){
3     super(props);
4     //code
5   }
6   loadPosts() {
7     //code
8   }
9   componentDidMount() {
10    //code
11  }
12 }

```

**Figure 4: componentDidMount() hook**

8. Implement the **render()** which will display the title and post of posts in html page using heading and paragraphs respectively.

```

JS Posts.js U X
1  class Posts extends React.Component {
2  >    constructor(props) { ...
5      }
6  >    loadPosts() { ...
8      }
9  >    componentDidMount() { ...
11     }
12     render() {
13         //code
14     }
15 }

```

**Figure 5: render() method**

9. Define a **componentDidCatch()** method which will be responsible for displaying any error happening in the component as alert messages.

```

JS Posts.js U X
1  class Posts extends React.Component {
2  >    constructor(props) { ...
5      }
6  >    loadPosts() { ...
8      }
9  >    componentDidMount() { ...
11     }
12 >    render() { ...
14     }
15     componentDidCatch(error, info) {
16         //code
17     }
18 }

```

**Figure 6: componentDidCatch() hook**

10. Add the Posts component to App component.
11. Build and Run the application using *npm start* command.

**Solution :**

**Posts.js**

```

class Post {
    constructor(id, title, body) {

```

```

    this.id = id;

    this.title = title;

    this.body = body;
  }
}

```

```
export default Post;
```

### **Post.js**

```

import React, { Component } from "react";
import Post from "./Post";

```

```
class Posts extends Component {
```

```

  constructor(props) {
    super(props);

```

```

    this.state = {
      posts: []
    };
  }

```

```
loadPosts() {
```

```

  fetch("https://jsonplaceholder.typicode.com/posts")
    .then(response => response.json())
    .then(data => {

```

```

    const postList = data.map(
      p => new Post(p.id, p.title, p.body)
    );

    this.setState({
      posts: postList
    });

  });
}

componentDidMount() {
  this.loadPosts();
}

componentDidCatch(error, info) {
  alert(error);
}

render() {

  return (

    <div>

```

```

    <h1>Posts</h1>

    {
      this.state.posts.map(post => {

        <div key={post.id}>

          <h2>{post.title}</h2>

          <p>{post.body}</p>

          <hr/>

        </div>

      })
    }

    </div>

  );

}

}

```

```
export default Posts;
```

### **App.js**

```
import Posts from "../Posts";
```

```
function App() {
```

```
  return (
```

```
    <div>
```

```
      <Posts />
```

```
    </div>
```

```
  );
```

```
}
```

```
export default App;
```





## Student Details:

**Name:** Steve

**School:** DNV Public School

**Total:** 284Marks

**Score:** 94.67%

### Exercise 5:

My Academy team at Cognizant want to create a dashboard containing the details of ongoing and completed cohorts. A react application is created which displays the detail of the cohorts using react component. You are assigned the task of styling these react components.

Download and build the attached react application.



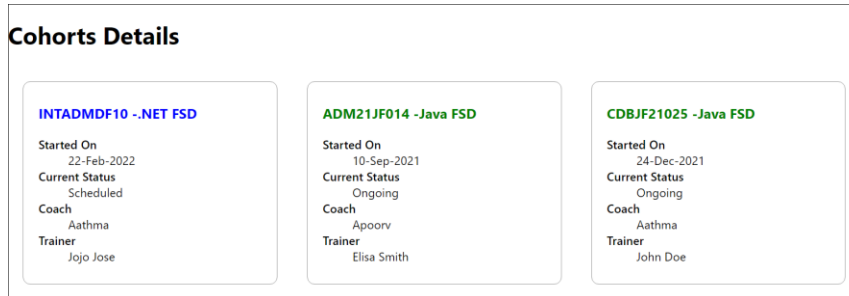
1. Unzip the react application in a folder
2. Open command prompt and switch to the react application folder
3. Restore the node packages using the following commands

```
C:\Windows\System32\cmd.exe  
C:\CTS-NewHandsOns\ReactHandsOns\cohortstracker>npm install
```

**Figure 7: Restore packages**

4. Open the application using VS Code
5. Create a new CSS Module in a file called "CohortDetails.module.css"
6. Define a css class with the name as "box" with following properties  
*Width = 300px;*  
*Display = inline block;*  
*Overall 10px margin*  
*Top and bottom padding as 10px*  
*Left and right padding as 20px*  
*1 px border in black color*  
*A border radius of 10px*
7. Define a css style for html <dt> element using tag selector. Set the font weight to 500.
8. Open the cohort details component and import the CSS Module
9. Apply the box class to the container div

10. Define the style for <h3> element to use “green” color font when cohort status is “ongoing” and “blue” color in all other scenarios.
11. Final result should look similar to the below image



**Figure 8: Final Result**

Solution :

### **CohortDetails.module.css**

```
.box{  
  
  width:300px;  
  
  display:inline-block;  
  
  margin:10px;  
  
  padding:10px 20px;  
  
  border:1px solid black;  
  
  border-radius:10px;  
  
}
```

```
dt{  
  
  font-weight:500;  
  
}
```

### **CohortDetails.js**

```
import styles from "./CohortDetails.module.css";
```

```

function CohortDetails(props) {

  return (

    <div className={styles.box}>

      <h3
        style={{
          color:
            props.status === "Ongoing"
              ? "green"
              : "blue"
        }}
      >

        {props.name}

      </h3>

      <dl>

        <dt>Started On</dt>
        <dd>{props.startDate}</dd>

        <dt>Current Status</dt>
        <dd>{props.status}</dd>

```

```

        <dt>Coach</dt>
        <dd>{props.coach}</dd>

        <dt>Trainer</dt>
        <dd>{props.trainer}</dd>

    </dl>

</div>

);

}

export default CohortDetails;

App.js
import CohortDetails from "./CohortDetails";
function App() {

    return (

        <div>

            <h1>Cohorts Details</h1>

            <CohortDetails

```

```

        name="INTADMDF10 - .NET FSD"
        startDate="22-Feb-2022"
        status="Scheduled"
        coach="Ashmita"
        trainer="Jojo Jone"
    />

    <CohortDetails
        name="ADM21JF014 - Java FSD"
        startDate="10-Sep-2021"
        status="Ongoing"
        coach="Apoorv"
        trainer="Elisa Smith"
    />

    <CohortDetails
        name="CDBJF21035 - Java FSD"
        startDate="24-Dec-2021"
        status="Ongoing"
        coach="Ashmita"
        trainer="John Doe"
    />
</div>

);
}

export default App;

```